Exp No: 9	INTELLIGENT TOURIST GUIDE CHATBOT
Date:	

PROBLEM TITLE:

TOURIST_CHATBOT – An Intelligent Travel Assistant for Tourist Guidance and Local Recommendations

PROBLEM STATEMENT:

- Tourists visiting new destinations often need real-time information about attractions, hotels, restaurants, transportation, and local activities. However, they may not always have access to reliable travel information or face language barriers while seeking assistance.
- The objective of this project is to develop a TOURIST_CHATBOT using Python that can interact with tourists, understand their travel preferences, provide personalized recommendations, and offer comprehensive local information without requiring external APIs or internet connectivity.
- The chatbot is designed for multiple travel categories including historical sites, adventure activities, dining experiences, accommodation options, and transportation services. The system maintains a knowledge base of popular destinations and provides assistance through text-based conversation using pattern matching and rule-based responses.

SOLUTION:

The proposed solution is TOURIST_CHATBOT, a Python-based intelligent travel guide system using a knowledge base and pattern matching without external APIs.

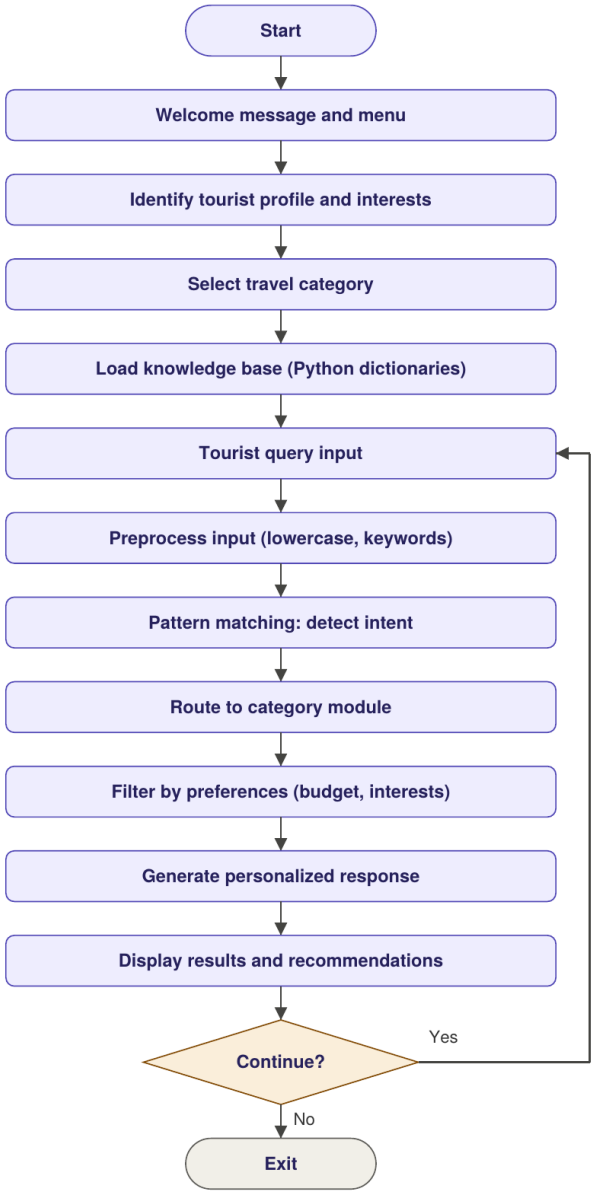
The system first identifies the tourist profile and travel interests. Based on preferences, the chatbot loads a suitable knowledge base containing information about attractions, hotels, restaurants, transportation, and local events.

The tourist can then:

- Search for popular tourist attractions and historical sites
- Get information about local restaurants and dining options
- Discover accommodation options with price ranges and facilities
- Learn about transportation methods and travel routes
- Receive personalized recommendations based on interests
- Access information about local events and cultural activities
- Get tips about safety, weather, and best travel times
- Plan itineraries based on available time and budget
- Find contact information and directions to destinations
- Get help with multilingual assistance and FAQ

The chatbot uses Python dictionaries for destination database, functions for query processing, conditional statements for preference filtering, pattern matching for input recognition, and string manipulation for response generation. No external API or database connection is required.

BLOCK DIAGRAM:



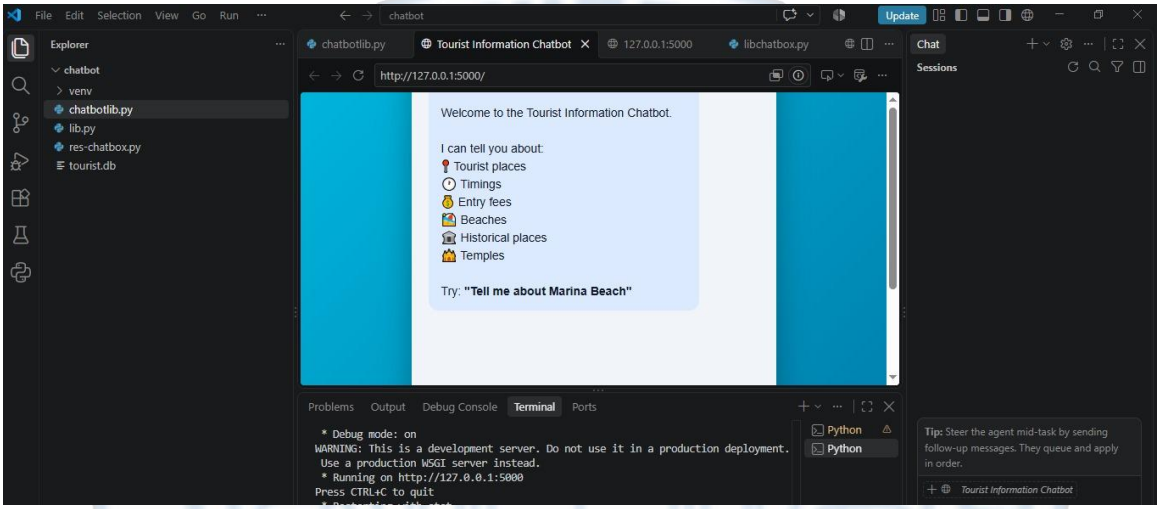
KEY FEATURES:

1. Attraction Information Module: Provides detailed information about tourist attractions, historical sites, museums, parks, and landmarks with descriptions, opening hours, and entry fees.
2. Accommodation Finder: Helps tourists find suitable hotels, hostels, and accommodations based on budget, location preferences, and facility requirements.
3. Dining Guide: Recommends restaurants, cafes, and local cuisines based on preferences, price range, and dietary restrictions.
4. Transportation Assistant: Provides information about local transit options, taxi services, rental options, and route planning.

5. Itinerary Planner: Helps create day-wise travel plans based on time availability, interests, and proximity of attractions.
6. Event Calendar: Displays upcoming events, festivals, cultural programs, and special activities happening during the visit.
7. Travel Tips and Safety: Provides practical information about weather, best seasons, safety precautions, and local customs.

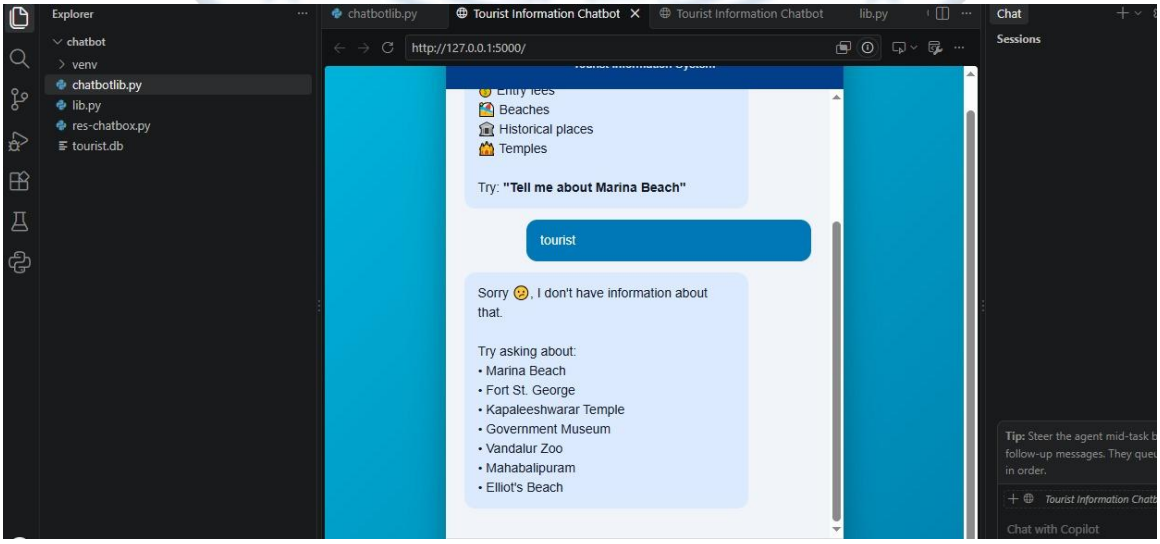
OUTPUT:

Figure: 1



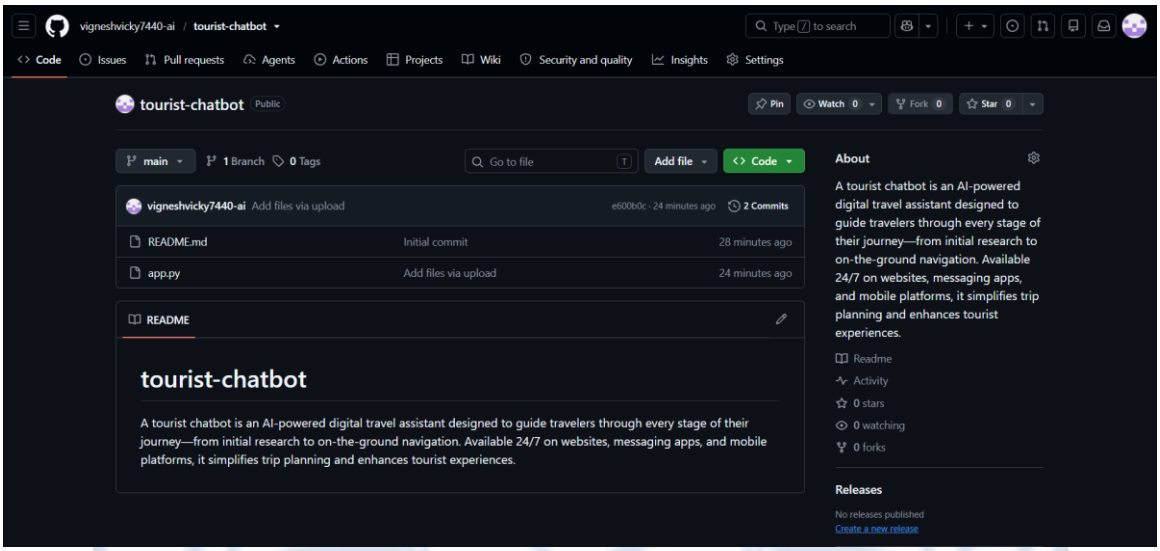
Explanation: The output section can display the main interface of the TOURIST_CHATBOT, showing a welcome message, tourist profile selection, destination/travel type selection, and a text input area for travel queries.

Figure: 2



Explanation: The chatbot processes a tourist query and provides relevant attraction, accommodation, dining, transportation, itinerary, or travel-tip information from its knowledge base. It can also provide personalized recommendations based on the tourist's selected preferences.

GITHUB:



OUTCOME:

The project successfully demonstrates the practical application of AI techniques for tourism automation without using Machine Learning or Deep Learning models. It uses rule-based pattern matching and knowledge base management to provide intelligent travel guidance.

Key Outcomes:

- Successfully implemented pattern-based query processing for travel information
- Efficiently searches and retrieves destination information from knowledge base
- Provides accurate personalized recommendations based on tourist preferences
- Reduces planning time and improves travel experience for tourists
- Provides a conversational and user-friendly travel guide interface
- Demonstrates practical AI without deep learning concepts
- Scalable design for expanding destination and service database

Exp No: 10	MINI-PROJECT
Date:	

PROBLEM TITLE:

AI-Based Hand Cricket Game with Intelligent Decision Making and Scoring System

PROBLEM STATEMENT:

- Hand cricket is a popular playground game that relies on quick decision-making and strategic thinking. Developing a digital version of this game requires implementing game rules, turn management, scoring logic, and an AI opponent that can make intelligent moves and adapt to player strategies.
- The proposed Hand Cricket Game provides an interactive digital implementation of the classic hand game with a playable AI opponent, realistic game mechanics, and engaging gameplay experience.

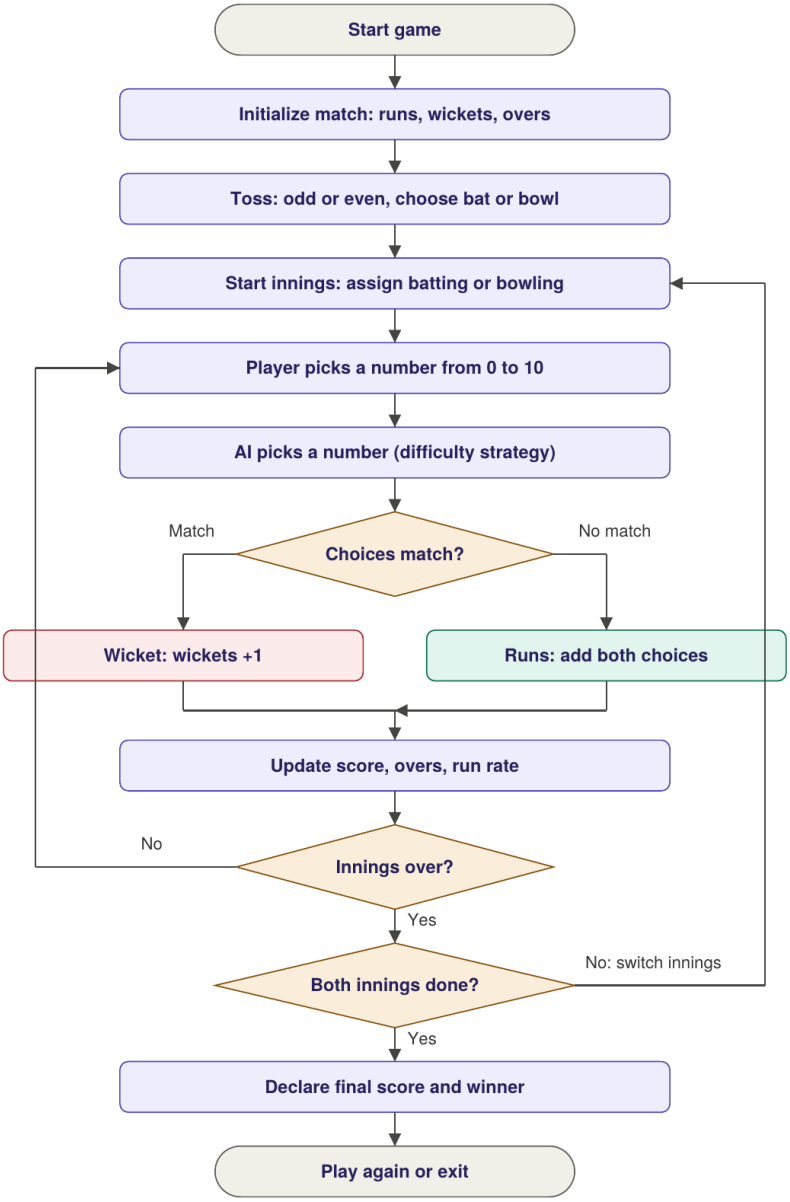
SOLUTION:

The proposed system implements hand cricket game logic with turn-based mechanics, scoring system, and an intelligent AI opponent using decision-making algorithms.

- Player chooses hand gesture (0-10 fingers) as batting move
- AI opponent simultaneously chooses its gesture
- Comparison logic determines run score (sum of both choices)
- Wicket logic determines when batter is out (matching choices)
- Turn-based innings system for batting and bowling
- AI opponent uses strategic decision-making for move selection
- AI adapts strategy based on player patterns and score
- Tracks runs, wickets, and overs for complete innings
- Implements cricket rules including innings end conditions
- Calculates final scores and determines match winner
- Displays statistics and performance metrics
- Provides replay functionality and match summary

This makes the game strategic, interactive, and demonstrates the practical use of game logic and AI decision-making in game development.

BLOCK DIAGRAM:



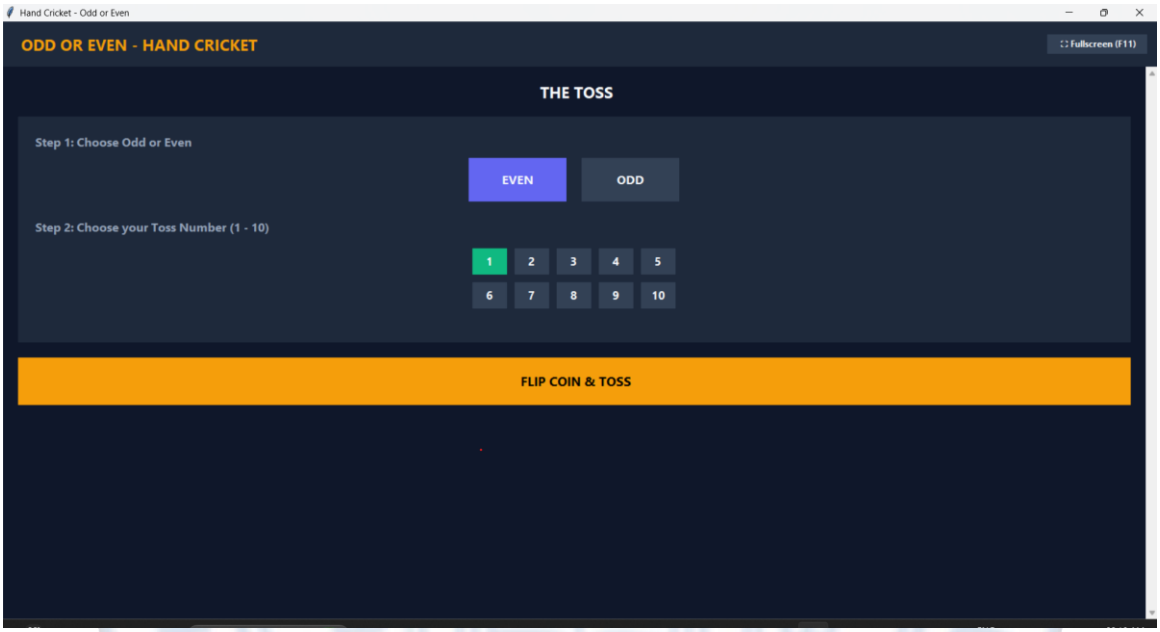
KEY FEATURES:

8. Game Mechanics: Implements hand cricket rules where each move is a choice of 0-10 fingers. Runs are calculated based on sum; wicket occurs on matching numbers.
9. Scoring System: Tracks runs per over, total score, wickets lost, and overs bowled with proper cricket statistics.
10. AI Opponent: Implements intelligent decision-making that analyzes player patterns, adapts strategies, and makes strategic moves.
11. Turn Management: Manages alternating player and AI turns, innings transitions, and match flow.
12. Match Statistics: Displays comprehensive statistics including run rate, wicket information, and performance analysis.

13. Win Conditions: Calculates match winner based on runs scored and properly implements cricket match rules.
14. Difficulty Levels: Multiple AI difficulty levels (Easy/Medium/Hard) affecting AI decision-making strategy.

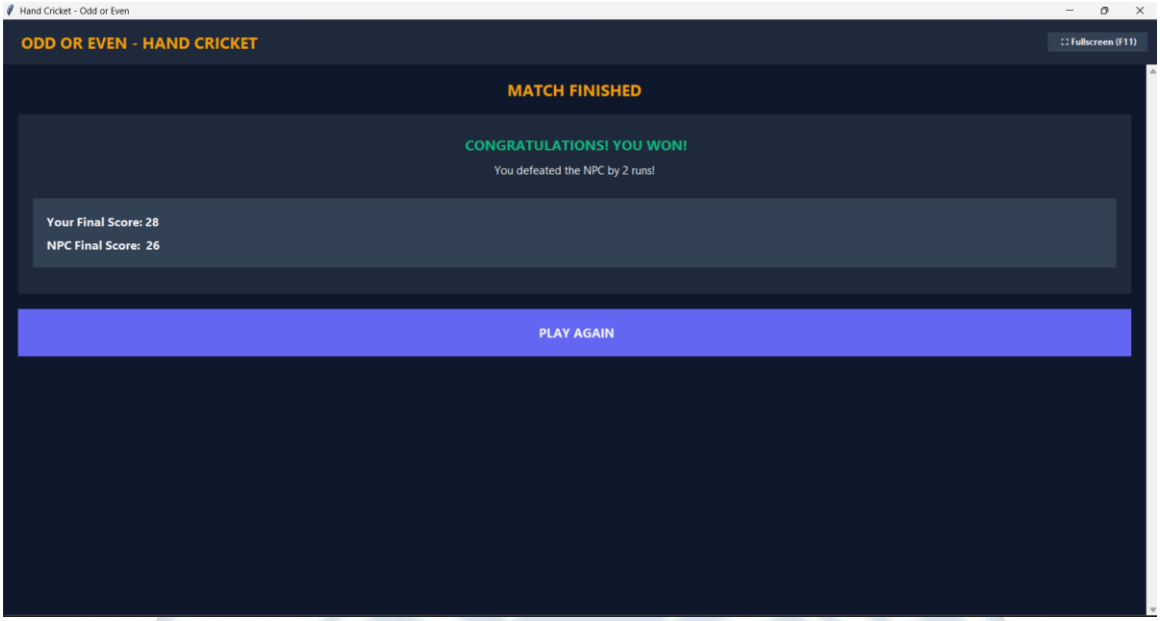
OUTPUT:

Figure: 1



Explanation: The output screen can show the Hand Cricket gameplay interface with player and AI choices, current innings, runs, wickets, overs, and game status.

Figure: 2



Explanation: The result screen displays the final score, winner determination, match statistics, and a Play Again/Replay option after the match ends.

Technologies Used

Programming Language: Python

Game Logic: Turn-based game mechanics

AI Technique: Decision-making and adaptive strategy

Game Environment: Digital hand-cricket interface

GITHUB:

OUTCOME:

The project successfully demonstrates game logic implementation and AI decision-making algorithms. The system uses mathematical calculations for scoring, conditional logic for game rules, and strategic algorithms for AI opponent behavior.

Key Outcomes:

- Successfully implemented hand cricket game rules and mechanics
- Efficient turn-based game system with proper state management
- Intelligent AI opponent with adaptive strategy selection
- Accurate scoring and wicket logic implementation
- Interactive gameplay with real-time feedback and statistics
- Demonstrates game development concepts without complex framework